

MIGOS 2026/27: Derived manifolds

Learning seminar for the CRC 1785

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Goals

One primary source of difficulties in the classical theory of manifolds are the singularities created by nontransverse intersections. The theory of derived manifolds provides a solution to this problem: it enlarges the class of smooth geometric objects to allow for well-behaved nontransverse intersections, preserving the transverse ones among manifolds.

In recent years, it is becoming clear that derived manifolds are a natural home for various moduli problems in geometry. One of the main goals of this seminar is to understand Steffens's theorem on derived representability for elliptic moduli problems, which says that the moduli space of solutions to an elliptic PDE is locally given by derived manifolds. Such moduli spaces are important in much of modern geometric topology, such as symplectic topology, Floer theory, complex geometry and the theory of 4-manifolds.

The seminar is intended to be accessible to first-year PhD students joining the CRC. In an attempt to keep the seminar as self-contained and self-motivated as possible, we devote a large part of the seminar to learning the background and tools from derived smooth geometry. Along the way, we will also encounter various higher categorical methods that make this theory work well, including the concepts of (homotopy) pullbacks in ∞ -categories, derived ∞ -categories, and (higher) stacks. Given the time limitations, these topics necessarily receive shorter treatment than they deserve, but we hope to illustrate why they are indispensable tools for the research of the CRC.

Each talk lasts 90 minutes, including questions from the audience.

Program

Talk 1, 13 October: Why derived manifolds?

Aim. Motivate derived manifolds through nontransverse intersections and elliptic moduli problems

Content. Prove that pullbacks of transverse intersection exist in the category of manifolds. Discuss the issues with non-transverse intersection. As a running example, compare a transverse intersection of two lines with a tangential intersection between a line and a parabola. Using more examples, explain how remembering the differential as a two-term complex solves some, but not all, problems. This shows that the correct home for such

intersections is necessarily “derived”. Give an informal discussion of the universal property of derived manifolds as a universal enlargement preserving transverse intersections.

End with giving a brief idea of moduli spaces of elliptic equations. Discuss the local finite-dimensional reduction of an elliptic equation, which allows one to present such moduli spaces locally as non-transverse intersections. Formulate the nonuniqueness and coherence problem for the resulting local presentations, again motivating derived manifolds.

Primary references. Spivak [Spi10, Section 1]; Steffens [Ste22, Chapter 1, especially Section 1.0.1]; Pardon [Par24, Sections 1–2].

Talk 2, 20 October: Introduction to C^∞ -rings

Aim. Encode smooth manifolds through their rings of smooth functions.

Content. Define C^∞ -rings and give their description as functors. Prove that $C^\infty(\mathbb{R}^n)$ is free on n generators. Prove Hadamard’s lemma and discuss quotients of C^∞ -rings. Define $C^\infty(X)$ for $X \subset \mathbb{R}^n$ an arbitrary closed subset. Prove that the resulting functor from closed subsets of euclidean spaces to C^∞ -rings is fully faithful.

Primary references. Moerdijk and Reyes [MR91, Chapter I, Section 1].

Talk 3, 27 October: Geometric constructions in C^∞ -rings

Aim. Understand the algebraic analogues of products and transverse pullbacks of manifolds.

Content. Introduce coproducts of C^∞ -rings. Prove Lemma 2.1 in Moerdijk–Reyes, and deduce that $C^\infty(M)$ is finitely presented for every manifold M . Prove that $C^\infty(M \times N) \simeq C^\infty(M) \otimes C^\infty(N)$. Prove that transverse pullbacks in manifolds induce pushouts of C^∞ -rings.

Primary references. Moerdijk and Reyes [MR91, Chapter I, Section 2].

Talk 4, 3 November: Derived manifolds

Aim. Understand the universal property of the ∞ -category of derived manifolds.

Content. Formulate the 1-categorical universal property of finitely presented C^∞ -rings. Use self-intersections to illustrate why ordinary pullbacks lose geometric information. Briefly introduce the idea behind ∞ -categories and homotopy pullbacks, without going into a model. Define derived manifolds via the ∞ -categorical analogue of the previous universal property. State the theorem of Carchedi–Steffens identifying them with finitely presented derived C^∞ -rings, discussing as much of the proof as time permits.

Primary references. Carchedi and Steffens [CS19].

Talk 5, 10 November: Calculating derived intersections

Aim. Learn to calculate derived zero loci via explicit dg-algebra models

Content. Introduce C^∞ -dg-algebras, state the C^∞ -Dold–Kan correspondence, and explain how it identifies derived zero loci with homotopy pushouts of C^∞ -dg-algebras. Construct the Koszul model for the derived zero locus of a smooth map $f: \mathbb{R}^n \rightarrow \mathbb{R}^r$ by resolving the evaluation map $C^\infty(\mathbb{R}^r) \rightarrow \mathbb{R}$. Explain the extension to families. Give examples, such as Spivak’s Example 2.7 and the zero loci of the equations x , x^2 , and 0, distinguishing the underlying zero set, classical truncation, and higher homotopy groups of the animated ring of functions.

Primary references. Steffens [Ste23, Definition 3.3.1, Theorem 3.3.12, and Examples 3.3.16–3.3.17]; Spivak [Spi10, Example 2.7].

Talk 6, 17 November: Tangent and cotangent complexes

Aim. Learn about (co)tangent complexes of derived manifolds.

Content. Define C^∞ -derivations, the module of smooth Kähler differentials, and its relation to the cotangent bundle of a manifold. Explain the need for a cotangent *complex* in the derived setting. Introduce the derived ∞ -category of real vector spaces, and discuss shifts and fiber/cofiber sequences, making the analogy to short exact sequences in abelian categories. Introduce derived square-zero extensions and C^∞ -derivations, and define absolute and relative cotangent complexes. State base change and the transitivity cofiber sequence for cotangent complexes, and use that to calculate the tangent complex of the derived zero locus of a section $s: U \rightarrow E$ at a zero x :

$$[T_x U \xrightarrow{d_x s} E_x].$$

Interpret its homology in degrees 0 and -1 as deformation and obstruction spaces.

Primary references. Steffens [Ste22, Section 5]; Pascarella [Pas25]; Spivak [Spi10, Section 7].

Talk 7, 24 November: Quasi-smooth derived geometry and Kuranishi charts

Aim. Understand quasi-smooth derived C^∞ -schemes and their local descriptions via Kuranishi charts.

Content. Introduce localizations of derived C^∞ -rings, stating flatness as a black-box, and define the spectrum of a derived C^∞ -ring. Define derived C^∞ -schemes locally of finite presentation, and identify the affine objects with derived manifolds. Define quasi-smooth derived C^∞ -schemes. Introduce Kuranishi charts (U, E, s) and their associated derived manifolds, and discuss the normal-form theorem: every quasi-smooth derived C^∞ -scheme of finite presentation is locally presented by a Kuranishi chart. Illustrate the non-uniqueness of the presentation by discussing stabilization of charts.

Primary references. Steffens [Ste22, Chapter 1 and Sections 4.1.3 and 5.1.1].

Talk 8, 1 December: Derived bordism and fundamental classes

Aim. Understand the benefits of derived bordism over classical bordism.

Content. Following Spivak [Spi10], explain how quasi-smooth derived manifolds are good for doing intersection theory. Define derived cobordism over X . State proposition 3.3, recalling its proof for derived C^∞ -rings. State proposition 3.4. Prove that the comparison map

$$\Omega(X) \xrightarrow{\cong} \Omega^{\text{der}}(X).$$

is an isomorphism by transverse perturbation. Deduce the intersection formula

$$[A] \smile [B] = [A \times_M^{\text{der}} B]$$

without transversality hypothesis. Discuss the derived self-intersection of the zero section of a vector bundle as the main example, recovering its Euler class.

Primary references. Spivak [Spi10, Sections 1–3].

Talk 9, 8 December: Stacks and local representability

Aim. Understand how stacks describe moduli problems.

Content. Define open coverings of derived manifolds and the resulting notion of (∞) -stacks on the category of derived manifolds via higher categorical descent. Briefly explain the comparison with smooth stacks. Define when such a derived C^∞ -stack is representable by a derived C^∞ -scheme, giving the derived zero locus of a section as a running example. Define effective epimorphisms and atlases. Prove the local criterion for representability, and explain how an open atlas produces a global representing derived C^∞ -scheme. Introduce relative representability and its locality on the base.

Primary references. Steffens [Ste24, Section 2.1].

Talk 10, 15 December: Section stacks and derived solution stacks

Aim. Construct the moduli stack of solutions to a differential operator

Content. Motivate differential operators and their solutions (e.g. harmonic functions, critical points of nonlinear energies, pseudo-holomorphic curves), and explain how to see these as solutions for $P(u) = b$ for a map $P: \Gamma_N(X \rightarrow N) \rightarrow \Gamma_N(Y \rightarrow N)$ and a section b . Introduce jet stacks to define finite-order differential operators and their induced maps on sections. Explain the relative version for smooth families $M \rightarrow S$, and briefly discuss how descent allows S to be a stack. Define differential moduli problems and their solution stacks. Explain linearization and *elliptic* differential moduli problems. As a running example, use

$$P(u) = \Delta u + u^2$$

on a closed connected Riemannian manifold.

Primary references. Steffens [Ste24, Definition 3.2.1, Definition 3.3.7, Proposition 3.3.9, and Definitions 3.4.1–3.4.2]; Pardon [Par24, Section 4].

Talk 11, 12 January: Elliptic representability I: local reductions

Aim. Reduce the representability problem to section stacks of vector bundles

Content. State the seminar’s main theorem: Steffens’s relative elliptic representability theorem [Ste24, Theorem 3.4.3]. For most of the talk, fix $S = \text{pt}$ and consider a moduli problem of the form $P(u) = b$ for a finite-order elliptic differential operator P and a section b . Use relative exponential maps and Lemma 3.4.4 to identify its solution stack locally around a solution u_0 with the solution stack of a moduli problem of the form $\tilde{P}(u) = 0$ for $\tilde{P}: Q \rightarrow \text{Map}_N(N, E)$, where $E, F \rightarrow N$ are vector bundles and $Q \subseteq \text{Map}_N(N, F)$ is an open substack. Explain how varying the chosen solution gives an effective open cover, so that local representability implies representability. Next, explain how the same coordinate construction reduces a moduli problem $P_1(u_1) = P_2(u_2)$ to this form. Finally, explain what changes when the base S is a general manifold or a smooth stack.

Primary references. Steffens [Ste24, Theorem 3.4.3, Proposition 3.2.19, Lemmas 3.2.20–3.2.21, Proposition 3.3.9, and Lemma 3.4.4]; Pardon [Par24, Introduction and the beginning of Section 5].

Talk 12, 19 January: Elliptic representability II: the local Kuranishi criterion

Aim. Explain the formal thickening argument that produces Kuranishi charts.

Content. Continue with $\tilde{P}(u) = 0$ over $S = \text{pt}$. For a finite-dimensional vector space V and a linear map $\iota: V \rightarrow \Gamma(N; E)$, introduce the augmented operator $\overline{P}(u, v) := \tilde{P}(u) + \iota(v)$. Show that projection to V identifies the original solution stack with the derived pullback

$$\text{Sol}(\tilde{P}) \simeq \text{Sol}(\overline{P}) \times_V^{\text{der}} \{0\}.$$

Deduce that this gives a local Kuranishi presentation whenever the augmented solution stack is represented by a finite-dimensional manifold near the chosen solution.

Regard the Fréchet space of smooth sections as a smooth stack, and compare it to the derived section stack. Introduce submersive maps of stacks, and show that passing from smooth to derived stacks preserves pullbacks. Deduce that the augmented solution stack is a finite-dimensional manifold whenever the augmented operator is submersive on a suitable neighbourhood. Explain the relative version over S .

Primary references. Steffens [Ste24, Propositions 2.2.17–2.2.18 and Corollary 2.2.19, Propositions 3.1.7 and 3.1.9 and Lemma 3.2.11, the final pullback argument following the proof of claim (*) in Theorem 3.4.3, Proposition 2.1.48 and Corollary 2.1.49].

Talk 13, 26 January: Elliptic representability III: the analytic input

Aim. Use nonlinear elliptic Fredholm analysis to make the augmented operator into a submersive map of smooth stacks

Content. Start with $S = \text{pt}$. Introduce the Sobolev spaces of sections $H^\ell(E; N)$ and explain how a differential operator of order k extends to compatible maps

$$\tilde{P}_\ell: Q_{k+\ell} \longrightarrow H^\ell(E; N)$$

for sufficiently large ℓ . Define Fredholm operators, and state as an analytic black-box that the Sobolev extensions of a linear elliptic operator on a compact manifold are Fredholm.

Given a solution t of $\tilde{P}$, use Fredholmness to choose a finite-dimensional augmentation whose derivative is surjective near $(t, 0)$.

For an ordinary manifold Z and a smooth family $h: Z \rightarrow \Gamma(N; E)$, apply the Banach implicit function theorem to the augmented equation $\tilde{P}(u) + \iota(a) = h(z)$. Use linear and nonlinear elliptic regularity to show that the resulting solution manifold represents the pullback of smooth stacks and projects submersively to Z , and conclude the submersiveness of the augmented operator as required by the previous talk. Explain the relative version over S , and conclude the representability theorem. Illustrate the argument with $P(u) = \Delta u + u^2$ near $u = 0$.

Primary references. Steffens [Ste24, Lemmas 3.4.5–3.4.6, the analytic part of the proof of Theorem 3.4.3]

Talk 14, 2 February: Pseudo-holomorphic curves as a derived moduli problem

Aim. Apply the representability theorem to a key geometric application.

Content. Describe families of maps from smooth compact Riemann surfaces to an almost complex manifold and formulate the Cauchy–Riemann equation as a section of the bundle of complex-antilinear maps. Explain the issue with the differential moduli problem in Steffens’s Construction 4.0.1, and use local coordinates and a trivialization of the equation bundle to obtain an elliptic operator between sections of vector bundles. Identify its linearization and prove its ellipticity. Apply the representability theorem locally and use descent to obtain the derived moduli stack of pseudo-holomorphic curves. Write the relative deformation-obstruction complex and interpret its homology. Given time, briefly discuss compactification and applications to enumerative questions.

Primary references. Wendl [Wen14, Sections 2.1, 2.3–2.4, and 3.3–3.4]; Steffens [Ste24, Section 4].

References

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